DESIGN AND DEVELOP FOCUS GUARD, A PHYSICAL USB-BASED SYSTEM FOR BLOCKING DISTRACTING WEBSITES

BY

OMOARE BLESSING OLUWAPELUMI	**2024212070069**
OSHO OMOBOLANLE AISHAT	**20200702020034**
OYEBADEJO EMMANUEL OLUWATOSIN	**2024212070073**
OMIKUNLE YUSUF ADESHOLA	**2020232070042**

SUBMITTED TO

DEPARTMENT OF COMPUTER ENGINEERING TECHNOLOGY, FACULTY OF ENGINEERING, THE POLYTECHNIC, IBADAN.OYO STATE, NIGERIA.

IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF HIGHER NATIONAL DEPLOMA (HND) IN COMPUTER ENGINEERING

FEB, 2026

CERTIFICATION

It is hereby certify that the project "Design and Develop Focus Guard, A Physical USB-Based System for Blocking Distracting Websites" was carried out by

Omoare Blessing Oluwapelumi	2024212070069
Osho Omobolanle Aishat	20200702020034
Oyebadejo Emmanuel Oluwatosin	2024212070073
Omikunle Yusuf Adeshola	2020232070042

_____ _____

Engr. Mrs. G. O Fadiran Date
Supervisor

_____ _____

Engr. O. A Busari Date
HOD

ACKNOWLEDEMENTS

We deeply appreciate our beloved parents whose unwavering love, prayers, and support have been a solid foundation throughout this academic journey. Our profound thanks also go to our Head of Department, Engr. O. A. Busari, for his leadership and encouragement. We are grateful to our project supervisor, Engr. Mrs. G. O. Fadiran, for her dedicated guidance and thoughtful suggestions that shaped the direction of this work. We also acknowledge Engr. O. A. Mowemi and other faculty members whose dedication and input contributed immensely to our growth. Above all, we return all glory to the Almighty God for His divine help and direction throughout this project.

DEDICATION

This report is wholeheartedly dedicated to the Almighty God, whose grace, wisdom, and strength have guided us through every step of this academic journey. We also dedicate this work to our beloved parents, whose unwavering love, prayers, support, and sacrifices have been the foundation of our success. Their encouragement and belief in our potential have been a constant source of motivation throughout our educational pursuits.

Table of Contents

FRONT PAGE…….……i

CERTIFICATION ii

ACKNOWLEDEMENTS..... iii

DEDICATION iv

Table of Content ---------- v

List of Table vii

List of Figure..... viii

CHAPTER ONE 8

INTRODUCTION..... 8

1.1 Introduction..... 8

1.2 Problem Statement 9

1.3 Aim and Objectives..... 10

1.3.1 Aim 10

1.3.2 Objectives 10

1.4 Significance of the Study 10

1.5 Scope and Limitations.....11

CHAPTER TWO 12

LITERATURE REVIEW..... 12

2.1 Historical Background of the Study..... 12

2.2 Review of Previous Studies 14

2.2.1 Digital Distraction and Academic Performance 14

2.2.2 Software-Based Website Blocking Solutions 15

2.2.3 Commitment Devices in Behavioral Economics 17

2.2.4 Hardware-Based Access Control Systems 18

2.2.5 User Experience and Behavioral Design Considerations 19
2.3 Gaps in Existing Research and Rationale for the Current Study 21
CHAPTER THREE 24
3.0 METHODOLOGY 24
3.1 System Architecture and Block Diagram 24
3.2 System Operation Flowchart 26
3.3 Functions of Major Components and Technologies 26
3.3.1 Python Programming Language 26
3.3.2 PyUSB Library 27
3.3.3 System Hosts File Manipulation 27
3.3.4 GUI Framework (Tkinter/PyQt5) 27
3.4 Implementation Methodology 27
3.4.1 Development Environment Setup 27
3.4.2 USB Device Detection Implementation 28
3.4.3 Website Blocking Implementation 28
3.4.4 Testing and Validation 28
3.5 Principle of Operation 28
3.6 System Requirements 29
3.6.1 Hardware Requirements 29
3.6.2 Software Requirements 29
3.7 Development Tools and Libraries 29

List of Table

Table 3.1: Development Tools and Libraries ……. 29

List of Figure

Fig 3.1: System Block Diagram the Focus Guard system ---------- 23

Fig 3.2: Flowchart diagram the Focus Guard system ---------- 25

CHAPTER ONE

INTRODUCTION

1.1 Introduction

In today's digital age, computers and internet-enabled devices have become essential tools for learning, work, and communication. While these technologies provide numerous benefits, they also introduce significant challenges, particularly in the form of digital distractions. Social media platforms, video streaming websites, online games, and constant notifications often interrupt concentration and reduce productivity among students, professionals, and researchers. As a result, maintaining sustained focus during study or work sessions has become increasingly difficult.

Existing solutions for managing digital distractions are mostly software-based, such as browser extensions and mobile or desktop applications that block distracting websites. However, many of these solutions are easily bypassed by disabling the application, uninstalling it, or entering a password to override restrictions. This weakness reduces their effectiveness, especially for users who struggle with self-control and require stronger commitment mechanisms to maintain focus.

To address this problem, this project presents the design and development of Focus Guard, a physical USB-based focus enforcement system that blocks access to distracting websites. Focus Guard introduces a novel approach by combining hardware and software to enforce discipline. The system operates using a USB device that serves as a physical commitment key; once connected, it activates a desktop application that restricts access to selected categories of websites for a defined period. Unlike traditional systems, Focus Guard removes password-based overrides and relies on physical presence of the USB device to maintain blocking, making circumvention significantly more difficult.

By integrating system-level blocking techniques such as DNS filtering, host file modification, and firewall rules, Focus Guard ensures robust website restriction across the operating system. The system also provides real-time feedback, motivational blocking screens, and productivity analytics to encourage positive focus habits. This project aims to offer an effective, practical, and discipline-oriented solution to the growing problem of digital distraction, particularly in academic and professional environments

1.2 Problem Statement

The rapid expansion of internet-enabled computers and digital devices has led to increased exposure to online distractions such as social media, video streaming platforms, online gaming, and other non-essential websites. While the internet plays a vital role in communication, learning, and work, its uncontrolled use has become a significant challenge to focus, discipline, and productivity across various sectors of society.

i. Families and Homes face difficulties in guiding children toward disciplined study habits, as unrestricted internet access often diverts attention away from homework, reading, and meaningful learning activities.

ii. Primary, Secondary, and Tertiary Educational Institutions experience reduced academic effectiveness as students struggle to maintain concentration during studying, online classes, assignments, and examination preparation due to constant digital distractions.

iii. Workplaces and Industrial Environments are affected by non-work-related internet usage during official hours, leading to decreased employee focus, lower productivity, and reduced organizational efficiency.

iv. Libraries and Study Centers, which are intended to support quiet and focused academic engagement, are increasingly challenged by misuse of internet resources for entertainment purposes.

v. Examination and Assessment Centers face risks related to loss of concentration and exposure to unauthorized online content during computer-based tests and evaluations.

vi. Training and Skill Development Centers encounter challenges in keeping trainees fully engaged with instructional materials due to easy access to distracting websites.

vii. Co-working Spaces struggle to support sustained focus among freelancers and remote workers who operate in shared digital environments.

viii. Research Institutions and Laboratories are negatively impacted by digital interruptions during tasks that require high levels of accuracy, analysis, and sustained attention.

x. Public Computer Centers and Cyber Cafés often lack effective control mechanisms to enforce responsible and purposeful internet usage.

xiii. Remote Learning and Online Education Environments struggle to ensure discipline, accountability, and sustained focus, especially where learners operate without direct supervision.

1.3 Aim and Objectives

1.3.1 Aim

The aim of this project is to design and develop Focus Guard, a physical USB-based system for blocking distracting websites to enhance user focus and productivity.

1.3.2 Objectives

i. To design a USB-based hardware–software system for enforcing focus sessions.
ii. To implement system-level website blocking for selected distracting website categories.
iii. To eliminate password-based overrides to prevent easy circumvention of the system.
iv. To provide real-time feedback and basic analytics to support productive user behavior.

1.4 Significance of the Study

This study is significant as it introduces a physical commitment-based approach to managing digital distractions. By integrating a USB device with system-level website blocking, Focus Guard provides a more effective alternative to traditional software-only solutions. The system can help students and professionals improve concentration, reduce unproductive internet usage, and develop better focus habits, while also contributing to research in hardware–software integrated productivity systems.

1.5 Scope and Limitations

Scope

The project focuses on the design and implementation of a USB-based website-blocking system for desktop and laptop computers. It restricts access to distracting websites during predefined focus sessions using system-level control techniques. The system is intended for use in academic, professional, and personal study environments.

Limitations

The system is limited to computers that support USB devices and may require administrative privileges during installation. It does not control distractions from mobile devices or offline applications. The effectiveness of the system also depends on the availability and proper functioning of the USB device, and advanced users may still attempt complex methods to bypass restrictions.

CHAPTER TWO

2.0 LITERATURE REVIEW

2.1 Historical Background of the Study

The digital age has fundamentally transformed how individuals work, study, and interact with information. With the proliferation of internet-connected devices and the ubiquity of online platforms, modern society has experienced unprecedented access to information and communication tools. However, this technological advancement has introduced a paradoxical challenge: while digital tools enhance productivity and enable remote collaboration, they simultaneously serve as sources of persistent distraction that undermine focus, concentration, and overall cognitive performance.

According to research conducted by the Program for International Student Assessment in 2022, approximately two-thirds of students in the United States reported being distracted by digital devices during class time, with 54 percent indicating distraction from other students' device usage. The correlation between digital distraction and academic performance is statistically significant: students who experienced distraction from peers using digital devices scored an average of 15 points lower in mathematics across OECD countries compared to those who were rarely or never distracted. This evidence underscores the pervasive nature of digital distraction and its measurable impact on learning outcomes.

The issue extends beyond educational settings into professional environments. Research by Marotta and Acquisti (2017) demonstrated through randomized field experiments that students assigned to use website blockers during tedious, attention-demanding tasks exhibited significantly higher productivity levels compared to control groups. Specifically, participants whose favorite websites were pre-blocked completed substantially more work than those who had voluntary access to blocking tools but did not configure them. However, the study also revealed an important behavioral nuance: forced blocking was associated with reduced user satisfaction, suggesting that autonomy and voluntary commitment play critical roles in the effectiveness of productivity interventions.

Empirical evidence from behavioral economics has established that individuals consistently struggle with temporal inconsistency—the gap between present intentions and future actions.

Research by Bucciol, Houser, and Piovesan (2011, 2013) demonstrated that when people attempt to work while simultaneously resisting desired activities, they experience reduced productivity, make more errors, and produce lower-quality outcomes. This phenomenon, known as the temptation-productivity trade-off, highlights a fundamental challenge: the presence of accessible distractions during cognitively demanding tasks impairs performance even when individuals possess the motivation to remain focused.

The proliferation of social media platforms, streaming services, gaming websites, and other engaging online content has intensified this challenge. Modern digital environments are deliberately designed to capture and retain user attention through algorithmic personalization, push notifications, and variable reward schedules. Research published in the Proceedings of the National Academy of Sciences found that interruptions occurring as frequently as every 10.5 minutes significantly fragment cognitive processing and reduce task completion efficiency. The cumulative effect of these micro-interruptions creates a state of continuous partial attention, wherein individuals never fully engage with any single task, thereby compromising both the quality and efficiency of their work.

Furthermore, studies have consistently demonstrated that multitasking—particularly media multitasking—is associated with poorer academic performance and reduced cognitive control. Research by Rosen, Carrier, and Cheever (2013) found that students who frequently switched between academic tasks and digital distractions took significantly longer to complete assignments and achieved lower test scores. Similarly, investigations by Junco and Cotten (2012) showed that college students engaging in frequent social media use during lectures retained less information than their focused counterparts. These findings challenge the widely held belief that younger generations possess superior multitasking abilities, revealing instead that sustained attention to a single task remains the optimal approach for learning and productivity.

Against this backdrop of mounting evidence regarding digital distraction, there has been growing interest in technological solutions designed to help individuals regain control over their attention. Software-based website blockers have emerged as one category of intervention, offering users the ability to restrict access to distracting websites during designated focus periods. However, existing software solutions face a critical limitation: they can be easily bypassed or disabled by users experiencing momentary impulses, thereby undermining their effectiveness as commitment

devices. This vulnerability highlights the need for more robust, physically enforced mechanisms that create meaningful barriers to impulsive behavior.

The concept of commitment devices—voluntary restrictions that individuals impose on their future selves to align actions with long-term goals—has a rich foundation in behavioral economics. Research by Bryan, Karlan, and Nelson (2010) established that commitment devices function by adding costs to non-compliance, thereby making it more difficult for individuals to deviate from intended behaviors. These mechanisms leverage insights from hyperbolic discounting theory, which posits that people tend to overvalue immediate rewards relative to future benefits, leading to present-biased decision-making. By creating structures that raise the cost of impulsive choices, commitment devices help individuals overcome this bias and maintain adherence to their stated objectives.

Hence, the current study proposes the design and development of Focus Guard, a physical USB-based system for blocking distracting websites. Unlike software-only solutions, Focus Guard introduces a hardware component that must be physically inserted into a computer to activate website blocking functionality. By creating a tangible, physical barrier to accessing blocked content, the system enhances the commitment mechanism and reduces the likelihood of impulsive deactivation. This approach addresses the limitations of existing software blockers while maintaining user autonomy through voluntary activation. The system aligns with contemporary research on commitment devices, digital well-being, and productivity enhancement, positioning it as a practical tool for students, professionals, and anyone seeking to improve focus in an increasingly distracting digital environment.

2.2 Review of Previous Studies

2.2.1 Digital Distraction and Academic Performance

The relationship between digital device usage and academic performance has been extensively documented in educational research. A systematic review conducted by researchers and published in Educational Technology Research and Development (2025) synthesized 26 empirical studies examining digital distraction in educational contexts. The review identified three primary categories of distraction causes: technology factors accounting for 51.95 percent of instances,

personal needs representing 37.66 percent, and instructional environment factors comprising 10.39 percent. The consequences of digital distraction were categorized as personal performance issues in 66.67 percent of cases, ineffective classroom instruction in 23.33 percent, and problematic technology use in 10 percent of instances.

Research published in peer-reviewed journals has demonstrated significant negative correlations between in-class cell phone use and course grades. One study analyzing eight introductory science courses at a large public university found that in-class cell phone usage corresponded to a grade reduction of 0.36 points on a 4.0 scale. Notably, observation data revealed that students consistently underreported the frequency of their digital device use, suggesting a gap between self-perception and actual behavior. This finding indicates that interventions must account for the tendency of individuals to minimize or deny the extent of their distraction.

Furthermore, research examining the cognitive mechanisms underlying distraction has revealed that the human brain cannot effectively multitask in the manner commonly assumed. Studies utilizing functional magnetic resonance imaging have shown that decisions involving immediate rewards activate the limbic system associated with affective decision-making, while choices involving future rewards engage the lateral prefrontal cortex linked to planning and control. This neurological evidence supports the behavioral observation that sustained, single-task focus produces superior outcomes compared to divided attention across multiple concurrent activities.

The implications of these findings extend to the design of productivity interventions. Understanding that digital distraction impairs academic performance through multiple mechanisms—cognitive overload, interrupted workflow, reduced information retention, and decreased engagement—provides a foundation for developing targeted solutions. Effective interventions must address not only the technological availability of distractions but also the psychological factors that drive individuals toward those distractions despite awareness of negative consequences.

2.2.2 Software-Based Website Blocking Solutions

The landscape of website blocking tools has evolved significantly over the past decade, with numerous software applications emerging to address productivity challenges. These solutions generally fall into three categories: browser extensions, standalone applications, and system-level

blocking tools. Research analyzing the effectiveness of blocking software has yielded important insights into both their benefits and limitations.

A field study conducted by Mark, Gudith, and Klocke (2017) involving 32 information workers examined the impact of blocking software on workplace focus and productivity. Participants worked for five days in a baseline condition followed by five days with distraction-blocking software activated. The results demonstrated that participants assessed their productivity significantly higher and could maintain focus for significantly longer periods when blocking software was active. Interestingly, the study found that individuals most distracted by social media derived the greatest benefit from blocking interventions, suggesting that the effectiveness of such tools varies based on individual distraction patterns and self-control capabilities.

Popular blocking tools such as Freedom, Cold Turkey Blocker, StayFocusd, and FocusMe offer various features including scheduled blocking sessions, cross-device synchronization, and customizable blocklists. Freedom, for instance, provides the ability to block websites, mobile applications, and desktop applications across multiple devices simultaneously. These tools operate by modifying system files, browser settings, or network configurations to prevent access to specified websites during designated time periods.

However, software-based solutions face inherent vulnerabilities. As purely digital interventions, they can typically be disabled, uninstalled, or circumvented through various means—including using different browsers, booting into alternative operating systems, or simply clicking through override prompts. Research examining workplace website blocking policies found that despite 64 percent of organizations implementing firewall-based restrictions, 40 percent of employees reported successfully accessing restricted content, primarily through mobile devices or workarounds. This pattern of circumvention undermines the commitment mechanism that blocking tools are designed to provide.

The ease of bypass in software solutions points to a critical gap in current productivity tools. While these applications demonstrate effectiveness when users maintain voluntary compliance, they offer limited protection against momentary impulses and present-biased decision-making. This limitation motivates the exploration of hardware-based solutions that introduce additional physical barriers to circumvention, thereby strengthening the commitment mechanism and improving long-term adherence to productivity goals.

2.2.3 Commitment Devices in Behavioral Economics

Commitment devices represent a well-established concept in behavioral economics, defined as voluntary restrictions that individuals impose on their future choice sets to overcome intrapersonal conflicts between present and future selves. The theoretical foundation for commitment devices stems from models of time-inconsistent preferences, particularly hyperbolic discounting, which describes the tendency to disproportionately value immediate rewards over delayed benefits.

Foundational research by Ariely and Wertenbroch (2002) demonstrated that individuals recognize their self-control limitations and voluntarily choose to impose constraints on their future behavior. In their experiments on procrastination and deadlines, participants who set binding deadlines for themselves performed better than those given complete flexibility, even though the self-imposed deadlines were strictly enforced. This finding established that commitment devices can effectively improve outcomes when individuals have insight into their own behavioral tendencies.

A comprehensive review by Bryan, Karlan, and Nelson (2010) synthesized evidence on commitment devices across multiple domains including savings behavior, addiction management, and productivity enhancement. The authors distinguished between 'hard' commitment devices, which impose irreversible or costly-to-reverse constraints, and 'soft' commitment devices, which rely primarily on psychological mechanisms such as goal-setting or public declarations. Their analysis revealed that hard commitments generally produce stronger behavioral effects but may also carry risks if individuals miscalculate their future preferences or capabilities.

An important study by Giné, Karlan, and Zinman (2010) examined commitment contracts for smoking cessation in the Philippines. Participants could voluntarily open a savings account wherein their deposits would be forfeited if they failed to pass a urine test for nicotine after six months. The results showed that individuals who opted into this commitment device were approximately 30 percent more likely to quit smoking compared to control groups. This evidence demonstrates the real-world effectiveness of commitment mechanisms in promoting behavior change.

However, research has also identified limitations and potential pitfalls of commitment devices. A field experiment by John (2018) found that while 45 percent of low-income individuals voluntarily adopted a savings commitment product, 55 percent subsequently defaulted on their self-imposed

commitments, incurring financial penalties. The study suggested that individuals may choose commitments that are either too weak to overcome their self-control problems or too strong given unforeseen circumstances. This finding highlights the importance of allowing users flexibility in calibrating the strength of their commitments.

In the context of productivity and digital well-being, commitment devices take various forms. Software applications like StickK allow users to set goals with financial stakes, where failure to meet commitments results in donations to disliked charities. Other tools employ social accountability mechanisms, wherein users share their goals with friends or colleagues who monitor progress. Physical commitment devices include time-lock containers for smartphones and website blocking routers that restrict internet access at the network level.

The theoretical and empirical literature on commitment devices provides strong support for hardware-based productivity interventions. By creating physical barriers that increase the cost of circumvention, such devices strengthen the commitment mechanism beyond what software-only solutions can achieve. The challenge lies in designing systems that offer sufficient constraint to overcome impulsive behavior while maintaining adequate flexibility to accommodate legitimate needs and prevent user frustration.

2.2.4 Hardware-Based Access Control Systems

While hardware-based systems for blocking distracting websites remain relatively unexplored in academic literature, the broader domain of hardware access control provides relevant precedents and technical foundations. USB-based security devices have been extensively developed for purposes including data loss prevention, device authentication, and malware protection.

Enterprise-level USB control solutions such as Endpoint Protector, AccessPatrol, and Device Control Plus enable organizations to manage and restrict USB device usage across networked computers. These systems typically operate through endpoint agents that enforce policies based on device whitelisting, user permissions, and file transfer restrictions. The technical architecture involves driver-level integration, hardware identification through Vendor ID and Product ID, and encrypted policy enforcement to prevent tampering.

Research in cybersecurity has highlighted the challenges of USB-based threats, including BadUSB attacks wherein malicious firmware can reprogram USB devices to impersonate keyboards,

network adapters, or other peripherals. While these security concerns differ from productivity applications, the underlying technical principles—hardware identification, driver interaction, and system-level access control—inform the design of USB-based productivity tools.

Physical USB port blockers represent the most rudimentary form of hardware-based access control. These devices mechanically prevent USB connections through lock-and-key systems or permanent epoxy filling. While effective at blocking physical access, such solutions lack the flexibility and intelligence required for productivity applications, as they prevent all USB functionality rather than selectively controlling network or application access.

A more sophisticated approach involves USB data blockers, colloquially known as 'USB condoms,' which allow charging functionality while blocking data transfer capabilities. These devices employ passive hardware filtering to physically disconnect data pins while maintaining power connections. This model demonstrates how USB hardware can enforce specific access policies through physical circuit design rather than software logic.

The integration of USB devices with software applications offers a hybrid model combining the tamper-resistance of hardware with the flexibility of software configuration. Examples include hardware security keys for two-factor authentication, which store cryptographic credentials on physical devices that must be present for authentication to succeed. These systems demonstrate how USB hardware can serve as a physical 'proof of intent' that users must provide before accessing protected resources.

Drawing from these precedents, a USB-based website blocking system would combine hardware presence detection with software-level network filtering. The physical USB device serves as both an authentication token and a commitment mechanism: users must physically insert the device to activate blocking, and removing the device would require a deliberate physical action rather than a simple software click. This design leverages the psychological impact of physical actions while maintaining the configurability necessary for personalized productivity strategies.

2.2.5 User Experience and Behavioral Design Considerations

The success of productivity interventions depends not only on technical effectiveness but also on user acceptance, sustained engagement, and behavioral compatibility. Research in human-

computer interaction has established that systems perceived as overly restrictive or inconvenient often face low adoption rates or high abandonment rates despite demonstrated efficacy.

A study examining workplace website blocking policies found that while such policies can enhance security and reduce time-wasting, they often generate negative reactions from employees who perceive them as signals of distrust. The research revealed that 18 percent of employees attempted to circumvent firewalls specifically to complete work-related tasks, suggesting that overly broad blocking can impede legitimate productivity. This finding emphasizes the importance of user control and customization in blocking systems.

Interviews with users of productivity tools have revealed diverse preferences regarding blocking strategies. Some individuals prefer complete internet blocking for deep work sessions, while others require access to research resources and communication tools. Temporal preferences also vary: some users benefit from scheduled blocking sessions aligned with their work patterns, while others prefer on-demand activation tied to specific tasks. Effective systems must accommodate this heterogeneity through flexible configuration options.

The concept of friction in behavioral design suggests that small increases in effort required to perform an action can significantly reduce the frequency of that action. Research has shown that even minimal delays—such as a 10-second countdown before allowing access to distracting websites—can interrupt automatic behaviors and prompt more deliberate decision-making. Hardware-based systems inherently introduce friction through the physical requirement to remove a USB device, potentially leveraging this mechanism to discourage impulsive circumvention.

However, excessive friction can lead to user frustration and abandonment. Studies of habit formation indicate that sustainable behavior change requires balancing structure with autonomy. Systems that allow users to set their own rules, adjust configurations based on learning, and temporarily disable restrictions for legitimate needs demonstrate higher long-term adherence than rigidly enforced controls.

These considerations inform the design requirements for Focus Guard. The system should provide sufficient constraint to serve as an effective commitment device while preserving user agency through customizable blocklists, flexible activation mechanisms, and transparent operation. The physical nature of the USB device serves as a tangible reminder of user commitment, while

software components enable the personalization necessary for diverse use cases and individual preferences.

2.3 Gaps in Existing Research and Rationale for the Current Study

Despite extensive research on digital distraction, commitment devices, and productivity interventions, significant gaps remain in the literature regarding practical, hardware-based solutions for website blocking. Current software-based blocking tools, while demonstrating effectiveness under conditions of user compliance, suffer from a fundamental limitation: they can be easily disabled through simple software actions. This vulnerability undermines their function as commitment devices, as users experiencing momentary impulses can bypass restrictions without meaningful barriers.

The behavioral economics literature has established theoretical foundations for commitment devices and demonstrated their effectiveness in domains such as savings behavior and addiction management. However, applications of these principles to digital productivity tools remain limited, particularly regarding the integration of physical hardware components. Existing research has not systematically explored how physical USB-based systems might strengthen commitment mechanisms beyond software-only approaches.

Furthermore, while USB access control systems have been extensively developed for security purposes, their potential application to individual productivity enhancement has received minimal attention. The technical architecture for hardware-software integration exists, but its adaptation to voluntary, user-controlled website blocking represents a novel application domain.

The current study addresses these gaps by proposing the design and development of Focus Guard, a physical USB-based system for blocking distracting websites. The system makes several distinctive contributions:

I. Integration of hardware and software components to create a stronger commitment mechanism than existing software-only solutions

II. Introduction of physical friction to reduce impulsive circumvention while maintaining user control through voluntary activation

III. Customizable blocking policies that allow users to tailor restrictions to their individual needs and productivity patterns

IV. Cross-platform compatibility to support users across different operating systems and computing environments

V. Cost-effective implementation accessible to students, professionals, and individuals seeking productivity enhancement

By combining insights from behavioral economics, user experience design, and hardware-software integration, Focus Guard represents a practical innovation in productivity technology. The system aligns with contemporary research on commitment devices while addressing the specific challenges of digital distraction in educational and professional contexts. Through systematic design, implementation, and evaluation, this study contributes both practical tools and empirical insights to the growing literature on digital well-being and productivity enhancement.

The development of Focus Guard also responds to documented user frustrations with existing solutions. As research has shown that easily bypassed software blockers fail to provide adequate commitment mechanisms for users with self-control challenges, a hardware-based approach offers a promising alternative. The physical requirement to remove a USB device creates a deliberate pause that may interrupt automatic behaviors and support more intentional decision-making regarding internet usage.

Therefore, this study provides both theoretical contributions to understanding commitment mechanisms in productivity interventions and practical contributions through the design of a novel hardware-software system. The research addresses identified gaps in the literature while responding to real-world needs articulated by students, professionals, and individuals seeking to improve their focus and productivity in an increasingly distracting digital environment.

The Focus Guard system is designed as an integrated hardware–software solution that combines a physical USB device with a desktop application to provide robust website-blocking functionality. The architecture is based on a commitment-device framework, in which the physical presence of the USB device serves both as an authentication mechanism and as a tangible reminder of the user's commitment to productivity. The system operates across multiple layers—hardware detection, application logic, network filtering, and user interface—all working in concert to deliver reliable distraction blocking while maintaining user control and configurability.

CHAPTER THREE

3.0 METHODOLOGY

3.1 System Architecture and Block Diagram

The Focus Guard system is designed as an integrated hardware–software solution that combines a physical USB device with a desktop application to provide robust website-blocking functionality. The architecture is based on a commitment-device framework, in which the physical presence of the USB device serves both as an authentication mechanism and as a tangible reminder of the user's commitment to productivity. The system operates across multiple layers—hardware detection, application logic, network filtering, and user interface—all working in concert to deliver reliable distraction blocking while maintaining user control and configurability.

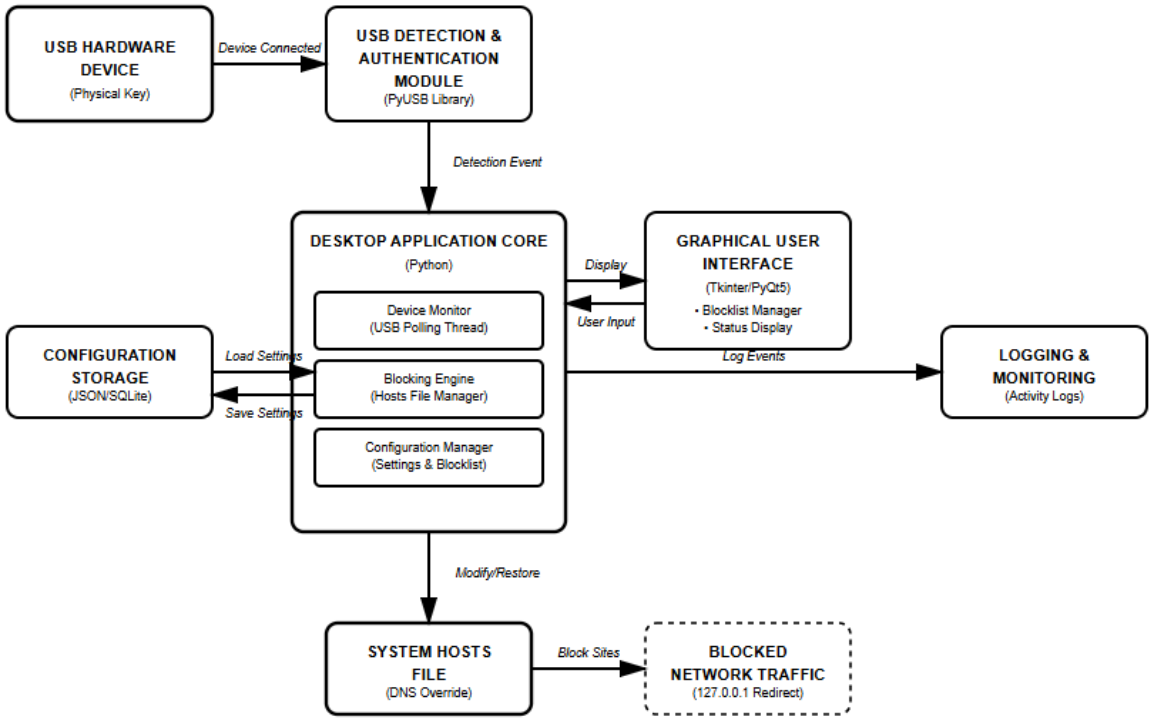

Fig 3.1: System Block Diagram the Focus Guard system

1. USB Hardware Device

The USB hardware device serves as the physical key that activates the Focus Guard blocking functionality. This device can be any standard USB flash drive or purpose-built USB token with a unique hardware identifier. When inserted into the computer, the operating system detects the device and makes it available to the Focus Guard application for verification through its Vendor ID (VID) and Product ID (PID). The physical nature of this component is critical to the system's effectiveness as a commitment device, as users must perform a deliberate physical action removing the USB device to deactivate blocking, rather than simply clicking a software button.

2. USB Detection and Authentication Module

The USB detection module continuously monitors the system for the presence of registered USB devices using the PyUSB library in Python. This component employs a polling mechanism that checks device presence at regular intervals (typically every 2-3 seconds) to detect insertion and removal events. The authentication mechanism verifies both the hardware identifiers and, optionally, a unique serial number stored in the device's firmware.

3. Desktop Application Core

The desktop application represents the central control unit of the Focus Guard system. Written entirely in Python, it coordinates all system operations including USB monitoring, configuration management, network filtering, and user interaction. The application runs as a background process with system tray integration, using object-oriented programming with distinct modules: DeviceMonitor for USB detection, BlockingEngine for website filtering, ConfigurationManager for settings, and UserInterface for graphical interaction through Tkinter or PyQt5.

4. Website Blocking Engine

The blocking engine implements website filtering through manipulation of the system hosts file. When activated, it reads the user-configured blocklist and modifies the hosts file to redirect blocked domains to localhost (127.0.0.1), preventing browsers from accessing those sites. This method works at the network layer, making it browser independent. The engine maintains a backup of the original hosts file for safe restoration when blocking is deactivated.

5. Configuration and Settings Manager

The configuration manager handles all persistent settings including the user's custom blocklist, registered USB device identifiers, blocking schedules, and application preferences. Settings are stored in a local JSON or SQLite database file, ensuring they persist across application restarts. Users can customize their blocklist, register multiple USB devices, and configure optional features through the graphical interface.

6. Graphical User Interface (GUI)

The GUI provides intuitive access to all Focus Guard features through a clean desktop application window built using Tkinter or PyQt5. It includes sections for blocklist management, USB device registration, real-time status monitoring, and settings configuration. System tray integration provides quick access without opening the full application window.

3.2 System Operation Flowchart

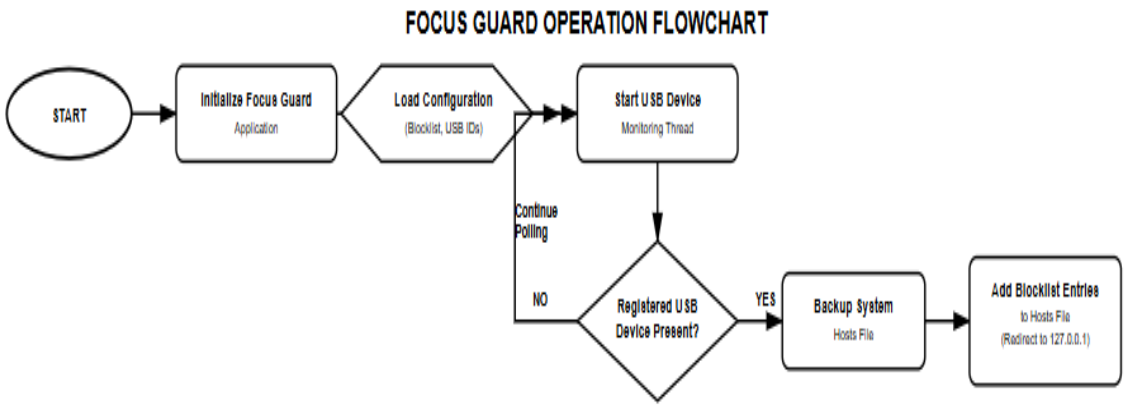

Fig 3.2: Flowchart diagram the Focus Guard system

3.3 Functions of Major Components and Technologies

3.3.1 Python Programming Language

Python serves as the primary development language for Focus Guard due to its extensive library support, cross-platform compatibility, and ease of development. Python's interpreted nature allows for rapid prototyping, while its robust standard library provides ready-made solutions for USB

device interaction, network configuration, GUI development, and file operations. Python 3.8 or higher is required for modern security features and library compatibility.

3.3.2 PyUSB Library

PyUSB is a Python wrapper for libusb, enabling USB device enumeration, device descriptor reading, and connection status monitoring. It abstracts low-level USB communication, providing a clean API for device detection. The library supports all major operating systems through platform-specific USB backends.

3.3.3 System Hosts File Manipulation

The hosts file maps domain names to IP addresses, overriding DNS resolution. By adding entries that redirect blocked domains to 127.0.0.1, the system prevents browsers from resolving those domains. The Focus Guard application requires administrator privileges to modify this file, implementing safe procedures including backup creation and atomic writes.

3.3.4 GUI Framework (Tkinter/PyQt5)

The graphical interface uses either Tkinter (lightweight, bundled with Python) or PyQt5 (feature-rich, professional appearance). Both frameworks support system tray integration, modal dialogs, and event-driven programming for responsive user interaction.

3.4 Implementation Methodology

The development of Focus Guard follows structured software engineering practices combining requirements analysis, system design, component implementation, integration testing, and deployment preparation.

3.4.1 Development Environment Setup

The development environment utilizes Python 3.8+ with a virtual environment for dependency isolation. Key dependencies include PyUSB, Tkinter/PyQt5, and platform-specific libraries. Development tools include Visual Studio Code or PyCharm IDE, Git for version control, and pytest for automated testing.

3.4.2 USB Device Detection Implementation

USB detection uses PyUSB's enumeration functions with a background monitoring thread polling at 2-second intervals. The system maintains a registry of authorized device identifiers in the configuration database. When a registered device is detected, callbacks trigger blocking activation; removal events trigger deactivation.

3.4.3 Website Blocking Implementation

Blocking is achieved through hosts file modification. The engine creates a backup, generates hosts file entries mapping blocked domains to 127.0.0.1, and uses atomic file operations for safe modification. Administrator privileges are requested on first run. Upon USB removal, the backup is restored.

3.4.4 Testing and Validation

Comprehensive testing includes unit tests for individual components, integration tests for component interaction, user acceptance testing, and cross-platform validation on Windows 10/11, macOS 12+, and Ubuntu Linux 20.04+.

3.5 Principle of Operation

Focus Guard operates on the principle of physical commitment enforcement combined with network-level website blocking. Upon startup, the USB monitoring module begins polling the USB subsystem using PyUSB to detect connected devices. Each USB device is identified by its unique Vendor ID and Product ID combination.

When a registered USB device is inserted, the blocking engine creates a backup of the system hosts file, then reads the blocklist and appends entries mapping each domain to 127.0.0.1. After modification, the DNS cache is flushed to ensure immediate blocking. The user interface displays confirmation of active blocking status.

While blocking is active, any attempts to access blocked websites fail as DNS resolution directs requests to localhost. This works independently of the browser or application being used. The monitoring module continues polling to detect USB device removal.

When the USB device is removed, the monitoring module triggers deactivation. The blocking engine restores the backed-up hosts file, flushes the DNS cache, and updates the user interface. The logging system records all events including activation/deactivation timestamps and configuration changes.

The physical USB device serves a critical psychological function as a commitment device. Unlike software-only blockers, Focus Guard requires deliberate physical action to deactivate, introducing meaningful friction that provides time for reflection and reduces impulsive decision-making.

3.6 System Requirements

3.6.1 Hardware Requirements

- Standard USB flash drive or USB device
- Computer with USB port (USB 2.0 or higher)
- Minimum 2GB RAM (4GB recommended)
- 100MB available disk space

3.6.2 Software Requirements

- Windows 10/11, macOS 12+, or Linux (Ubuntu 20.04+)
- Python 3.8+
- Administrator/root privileges

3.7 Development Tools and Libraries

Table 3.1: Development Tools and Libraries

Component	Technology/Library
Programming Language	Python 3.8+
USB Device Access	PyUSB 1.2.1, libusb 1.0
GUI Framework	Tkinter or PyQt5 5.15+

Configuration Storage	JSON or SQLite3
Testing Framework	pytest 7.0+, unittest
Packaging	PyInstaller 5.0+

This comprehensive set of tools and libraries ensures robust functionality, cross-platform compatibility, and maintainable code architecture for the Focus Guard system.